Red Hat OpenShift Data Foundation 4.22

4.22 Release Notes

Release notes for features and enhancements, known issues, and other important information.

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Legal Notice

Abstract

The release notes for Red Hat OpenShift Data Foundation 4.22 summarizes all new features and enhancements, notable technical changes, and any known bugs upon general availability.

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CHAPTER 1. OVERVIEW

Red Hat OpenShift Data Foundation is software-defined storage that is optimized for container environments. It runs as an operator on OpenShift Container Platform to provide highly integrated and simplified persistent storage management for containers.

Red Hat OpenShift Data Foundation is integrated into the latest Red Hat OpenShift Container Platform to address platform services, application portability, and persistence challenges. It provides a highly scalable backend for the next generation of cloud-native applications, built on a technology stack that includes Red Hat Ceph Storage, the Rook.io Operator, and NooBaa's Multicloud Object Gateway technology.

Red Hat OpenShift Data Foundation is designed for FIPS. When running on RHEL or RHEL CoreOS booted in FIPS mode, OpenShift Container Platform core components use the RHEL cryptographic libraries submitted to NIST for FIPS Validation on only the x86_64, ppc64le, and s390X architectures. For more information about the NIST validation program, see Cryptographic Module Validation Program. For the latest NIST status for the individual versions of the RHEL cryptographic libraries submitted for validation, see Compliance Activities and Government Standards.

Red Hat OpenShift Data Foundation provides a trusted, enterprise-grade application development environment that simplifies and enhances the user experience across the application lifecycle in a number of ways:

- Provides block storage for databases.
- Shared file storage for continuous integration, messaging, and data aggregation.
- Object storage for cloud-first development, archival, backup, and media storage.
- Scale applications and data exponentially.
- Attach and detach persistent data volumes at an accelerated rate.
- Stretch clusters across multiple data-centers or availability zones.
- Establish a comprehensive application container registry.
- Support the next generation of OpenShift workloads such as Data Analytics, Artificial Intelligence, Machine Learning, Deep Learning, and Internet of Things (IoT).
- Dynamically provision not only application containers, but data service volumes and containers, as well as additional OpenShift Container Platform nodes, Elastic Block Store (EBS) volumes and other infrastructure services.

1.1. ABOUT THIS RELEASE

Red Hat OpenShift Data Foundation 4.22 is now available. New enhancements, features, and known issues that pertain to OpenShift Data Foundation 4.22 are included in this topic.

Red Hat OpenShift Data Foundation 4.22 is supported on the Red Hat OpenShift Container Platform version 4.22. For more information, see Red Hat OpenShift Data Foundation Supportability and Interoperability Checker.

For Red Hat OpenShift Data Foundation life cycle information, refer Red Hat OpenShift Data Foundation Life Cycle.

CHAPTER 2. NEW FEATURES

This section describes new features introduced in Red Hat OpenShift Data Foundation 4.22.

2.1. SUPPORT FOR TWO NODES FENCING

OpenShift Data Foundation (ODF) on a two-node OpenShift cluster with fencing enabled is now supported. This architecture is suitable for distributed and edge deployments, where the goal is to minimize hardware footprint while maintaining High Availability (HA).

For more information, see Deploying OpenShift Data Foundation on two-node clusters .

2.2. ERASURE CODING SUPPORT IN INTERNAL MODE FOR RGW, RBD, CEPHFS

Erasure coding support for OpenShift Data Foundation (ODF) internal mode helps to enhance storage efficiency and reduce infrastructure costs. Erasure coding enables more space-efficient data protection compared to traditional replication, making it an ideal solution for customers seeking scalable and cost-effective storage.

For RADOS Gateway RGW object storage, erasure coding can only be enabled when creating the OpenShift Data Foundation cluster.

For RBD and CephFS storage, erasure coding can be enabled during cluster creation or later by creating a custom StorageClass.

For instructions on enabling erasure coding, refer to the deployment guide applicable to your environment.

2.3. SUPPORT FOR S3 VECTORS

S3 Vectors provide native vector-database storage capabilities within the Multicloud Object Gateway (MCG). Vector buckets enable you to store and manage vector embeddings for machine learning and AI applications, backed by NSFS (NamespaceStore FileSystem) storage.

For more information, see S3 Vectors in Multicloud Object Gateway .

CHAPTER 3. ENHANCEMENTS

This section describes the major enhancements introduced in Red Hat OpenShift Data foundation 4.22.

3.1. DISASTER RECOVERY

3.1.1. Transition to a Vendor-Agnostic DR Operator

The OpenShift Data Foundation Multicluster Orchestrator (ODF MCO) has been rebranded into a vendor-agnostic DR operator.

This update removes OpenShift Data Foundation specific branding and renames the operator to Data Foundation Multicluster Orchestrator.

3.1.2. Monitoring disaster recovery relationships using Topology view

The disaster recovery (DR) **Topology** view provides a visual, hierarchical representation of disaster recovery relationships across clusters and applications. It is designed to help users quickly understand DR coverage, health, and dependencies without navigating multiple pages or inspecting configuration details.

For more information, see Monitoring disaster recovery relationships using Topology view .

3.1.3. RDR for erasure coding (RBD and CephFS)

Erasure-coded (EC) enabled storage classes are supported for Regional Disaster Recovery (RDR). RDR between erasure-coded and non–erasure-coded storage classes is not supported.

3.1.4. Storage agnostic disaster recovery for third-party vendors

Ramen now supports an Agnostic DR framework that decouples workflows from Red Hat OpenShift Data Foundation and Ceph-backed storage. With this enhancement, Ramen abstracts the underlying storage layer and enables consistent disaster recovery operations across diverse infrastructures.

As long as a third-party storage vendor provides an operator capable of reconciling the VolumeGroupReplication Custom Resources (CRs), Ramen can orchestrate disaster recovery (DR) policies, unplanned failovers, and planned relocations in the same way, regardless of the storage platform.

For more information, see Storage agnostic disaster recovery for third-party vendors .

3.2. ENHANCED INFRASTRUCTURE HEALTH VISIBILITY

The OpenShift Data Foundation infrastructure health view has been upgraded to include the following:

- **New storage stackCeph alerts**: Added proactive notifications for critical underlying Ceph storage events to help you catch issues early.
- **Smart health filtering**: Use the new **All checks** dropdown to instantly filter health checks by their severity level or silence status.
- **Indefinite Alert Silencing**: You can now silence specific health checks indefinitely.

- **Interactive timeline charts**: Use new zoom and pan controls on the health score chart to isolate specific time ranges. This makes it easier to correlate historical infrastructure events with sudden changes in cluster health.

For more information, see Viewing OpenShift Data Foundation infrastructure health.

3.3. MULTICLOUD OBJECT GATEWAY

3.3.1. Object bucket quota with OBC

Multicloud Object Gateway (MCG) now supports quota configuration for Object Bucket Claims (OBC) and buckets to prevent resource starvation and optimize storage usage. Quotas help administrators control storage consumption and ensure fair resource allocation across multiple applications.

Quota configuration provides the following benefits:

- **Resource Management**: Prevents individual buckets from consuming excessive storage resources.
- **Cost Control**: Limits storage usage to control costs in cloud environments.
- **Multi-tenancy**: Ensures fair resource allocation across multiple applications and teams.
- **Capacity Planning**: Enables better prediction and management of storage requirements.

For more information, see Managing bucket quota.

3.3.2. Alert when MCG replication target is unreachable

There is a new health alert **NooBaaReplicationTargetUnreachable**. This means a NooBaa replication from a source bucket to a target bucket is failing for more than 5 minutes.

For more information, see NooBaaReplicationTargetUnreachable.

3.3.3. Support the standardized Azure Identity configuration flow via OLM and CCO for ODF

A standardized configuration workflow for Azure Identity on Azure Identity-enabled OpenShift clusters is supported. This enhancement provides a consistent and repeatable process using well-defined inputs and behavior, aligned with the common credential management flow. It simplifies deployment and reduces configuration complexity across operators that support Azure Identity. As a result, users can more easily adopt secure, token-based authentication with Azure services.

- **NooBaa operator log levels are now configurable to reduce log verbosity**
  The NooBaa operator log verbosity can now be adjusted to significantly reduce log spam during routine reconciliation loops. This has been resolved by introducing a configurable **OPERATOR_LOG_LEVEL** environment variable, allowing administrators to increase or reduce operator logging levels based on their operational needs.

  (DFBUGS-4372)

3.4. RADOS OBJECT GATEWAY (RGW)

3.4.1. STS authentication for RADOS Object Gateway by using CephObjectStore

Security Token Service (STS) token-based authentication for internal RADOS Object Gateway (RGW) instances is now enabled by default. This enhancement improves security and simplifies deployment by eliminating the need for manual STS configuration.

3.4.2. Disabling HTTP route for RADOS Object Gateway

You can disable insecure HTTP access to the RADOS Object Gateway (RGW) by configuring the disableHTTP option in the StorageCluster custom resource. This ensures that only secure HTTPS connections are allowed to access the RGW S3 endpoint, improving the security posture of your object storage deployment. You can disable HTTP route after the deployment of the cluster.

For more information, see Disabling HTTP route for RADOS Object Gateway .

3.5. VIEW CURRENT TOP 10 CEPHFS SUBVOLUMES ON ALL CLUSTERS

A new card has been added to the Block and File dashboard that shows the top 10 CephFS Persistent Volume Claims (PVCs) in order to identify potential bottlenecks in the storage cluster. You can pinpoint which volumes are driving the highest IOPS, experiencing the most latency, or consuming the most bandwidth (throughput).

For more information, see Metrics in the Block and File dashboard .

3.6. CLEANING ORPHANED CEPHFS SNAPSHOTS

Orphaned CephFS snapshots can accumulate over time when volume snapshots are deleted but their underlying CephFS snapshots remain. These orphaned snapshots consume storage space and can impact cluster performance. You can now identify and clean up orphaned CephFS snapshots using the ODF CLI tool.

For more information, see Cleaning orphaned CephFS snapshots.

3.7. MULTUS VALIDATION TOOL INTEGRATED INTO ODF-CLI

The Multus validation command that verifies the Multus and system configuration can now be run using the OpenShift Data Foundation CLI.

For more information, see Multus prerequisite validation.

3.8. ENABLE FORCEFUL DEPLOYMENT FROM THE OPENSHIT WEB CONSOLE INTERNAL MODE

There is now an Advanced setting to Enable forceful deployment when creating an OpenShift Data Foundation cluster in internal mode using local storage-based clusters. This setting is not applicable to clusters deployed in dynamic storage mode.

3.9. TOPOLOGY AWARENESS IN EXTERNAL MODE

Topology awareness ensures that data for persistent volumes (PVs) is provisioned from Ceph OSDs that belong to the same failure domain as the OpenShift node where the application pod runs.

For more information, see Configuring topology awareness for external mode.

3.10. CENTRALIZED CONFIGURATION FOR TLS PROFILES

This release allows configuring TLS servers protocol version, cipher suites, and groups across OpenShift Data Foundation in a centralized fashion. This update includes support for Post-Quantum hybrid key exchange mechanisms.

For more information, see Centralizing TLS security profiles for OpenShift Data Foundation .

3.11. STAGGERED SCHEDULE FOR RECLAIM SPACE JOBS

When any schedule for ReclaimSpace is used, including @daily, @weekly, @monthly, and cron expressions such as **0 * * * ***, multiple CronJobs are created. To prevent all Jobs from running simultaneously and overloading high-scale systems, executions are staggered within a 2-hour window from the scheduled start time (e.g., @weekly would be 00:00, 00:05, 00:10, and so on).

For more information, see Enabling reclaim space operation using ReclaimSpaceCronJob .

CHAPTER 4. TECHNOLOGY PREVIEW FEATURES

This section describes Technology Preview features introduced in Red Hat OpenShift Data Foundation 4.22.

IMPORTANT

Technology Preview features are subject to Technology Preview support limitations. Technology Preview features are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using them in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

For more information, see Technology Preview Features Support Scope .

4.1. DRYRUN MODE FOR TESTING FAILOVER

Testing Failover, also known as DryRun mode, allows you to validate disaster recovery procedures without disrupting your production application. This feature enables non-destructive failover testing by temporarily promoting the secondary cluster to verify readiness and data consistency while keeping the primary cluster completely untouched.

For more information, see DryRun mode for testing failover .

CHAPTER 5. DEVELOPER PREVIEW FEATURES

This section describes Developer Preview features introduced in Red Hat OpenShift Data Foundation 4.22.

IMPORTANT

Developer preview features are subject to Developer preview support limitations. Developer preview releases are not intended to be run in production environments. The clusters deployed with developer preview features are considered to be development clusters and are not supported through the Red Hat Customer Portal case management system. If you need assistance with developer preview features, reach out to the ocs-devpreview@redhat.com mailing list and a member of the Red Hat Development Team will assist you as quickly as possible based on availability and work schedules.

5.1. SERVER-SIDE ENCRYPTION FOR RGW (S3)

Introduces server-side encryption for S3 clients accessing RGW, with or without external key management systems (Vault, HPCS, KMIP). This capability enables more granular data protection by encrypting object data during runtime access, extending beyond existing encryption-at-rest mechanisms.

For more information, see SSE-S3 with Vault Agent for RGW.

5.2. GKLM KMS INTEGRATION FOR CLUSTER WIDE AND PV ENCRYPTION

Expands encryption key management capabilities through IBM Security Guardium Key Lifecycle Manager (GKLM) integration. This supports cluster-wide and persistent volume encryption while aligning with existing GKLM-based environments.

For more information, see How to use GKLM KMS with ODF .

5.3. CEPHFS CFT-BASED RDR REPLICATION

Enables CephFS replication leveraging Change File Tracking (CFT) to detect incremental changes between snapshots. This enables more efficient replication by identifying only modified files and changed blocks, reducing the need for full filesystem scans and improving performance in CephFS RDR scenarios.

For more information, see Enable Regional Disaster Recovery with CephFS Change File Tracking Based Replication in OpenShift Data Foundation 4.22.

5.4. CONFIGURABLE THRESHOLD FOR NOOBAA ALERTS

Adds capability allowing configurable thresholds for NooBaa alerts. This enables organizations to tailor alerting to their operational scale, helping improve preparedness and reduce the risk of capacity-related disruptions.

For more information, see Configuring NooBaa Prometheus Alert Rules .

CHAPTER 6. BUG FIXES

This section describes bugs fixed in Red Hat OpenShift Data Foundation 4.22.

6.1. MULTICLOUD OBJECT GATEWAY

- **noobaa-db-pg-cluster now synchronizes successfully without WAL segment errors**
  The **noobaa-db-pg-cluster** database cluster now synchronizes properly and avoids synchronization loops. Previously, the cluster would stop synchronizing and log an error indicating that a requested Write-Ahead Logging (WAL) segment had already been removed. To resolve this and increase replication resiliency, the configuration parameters **wal_keep_size**, **wal_level**, **wal_sender_timeout**, **wal_receiver_timeout**, and **failoverDelay** have been updated for NooBaa's PostgreSQL database usage.

  (DFBUGS-6943)

- **Intermittent S3 upload failures (HTTP 500) to NooBaa buckets have been resolved**
  Uploading files to NooBaa buckets via JFrog Artifactory no longer fails intermittently with **HTTP 500** / **InternalError**. Previously, concurrent **PUT** uploads to the same bucket and key could cause a database unique constraint conflict when attempting to update version records simultaneously, causing the second upload to fail. This issue has been resolved by implementing internal retry logic within the upload completion flow.

  (DFBUGS-6548)

- **NooBaa database cleaner environment variables are no longer overridden during reconciliation**
  Environment variables configured for the NooBaa database cleaner (**DB_CLEANER**) now properly persist and take effect. Previously, the **noobaa-operator** would overwrite custom environment variables on the core pod during its reconciliation loop, causing user-defined values (such as database cleaner timers) to be ignored. This issue has been resolved by ensuring the operator retains and respects existing custom configurations.

  (DFBUGS-5052)

- **The PodDisruptionBudgetAtLimit alert for the noobaa-db-pg-cluster-primary PDB is now automatically silenced**
  A misleading warning alert is no longer triggered for single-primary database configurations. Previously, CloudNativePG (the database operator used by NooBaa) created a PodDisruptionBudget (PDB) object to protect the primary database instance from deletion. Because there is only one primary, the PDB correctly allowed zero disruptions, which inadvertently caused OpenShift to fire a misleading **PodDisruptionBudgetAtLimit** warning alert. This issue has been resolved by updating the NooBaa operator to automatically silence this specific alert using the Alertmanager API.

  (DFBUGS-5294)

- **noobaa-core logs are no longer spammed by SignatureDoesNotMatch errors**
  S3 operations now succeed without signature validation failures, and the **noobaa-core** logs are no longer filled with error messages. Previously, a regression introduced during the migration to AWS SDK V3 caused incorrect signature calculations, which resulted in failed S3 operations and log spam. This issue has been resolved by removing **applyChecksum** from the requests.

  (DFBUGS-4461)

6.2. DISASTER RECOVERY

- **Regional-DR is now supported in environments deployed on IBM Power**
  Regional-DR is now supported in OpenShift Data Foundation environments deployed on IBM Power because ACM 2.15 support has been introduced for this platform in this release. This applies to both new and upgraded deployments on IBM Power.

  (DFBUGS-5369)

- **CIDR range does now persists incsiaddonsnode object when the respective node is down**
  The Classless Inter-Domain Routing (CIDR) information now correctly persists in the **csiaddonsnode** object even when a node is down. This ensures the fencing mechanism functions as intended when required to fence impacted nodes, eliminating the need to manually collect CIDR information immediately after the **NetworkFenceClass** object is created.

  (DFBUGS-2948)

- **Remove DR option is now available for discovered apps on the Virtual machines page**
  The **Remove DR** option is now available for discovered applications listed on the **Virtual machines** page. You no longer need to manually add the missing label to the **DRPlacementControl** or patch the **PROTECTED_VMS** recipe parameter as a workaround.

  (DFBUGS-2823)

- **DR Status is now displayed for discovered apps on the Virtual machines page**
  The DR Status is now correctly displayed for discovered applications listed on the **Virtual machines** page. Manually adding the missing label to the **DRPlacementControl** and patching the **PROTECTED_VMS** recipe parameter is no longer required.

  (DFBUGS-2822)

- **Secondary PVCs are now removed when DR protection is removed for discovered apps**
  On the secondary cluster, CephFS PVCs linked to a workload discovered using the Discovered Applications feature are now correctly marked as VRG-owned. As a result, when the workload is disabled, these PVCs are automatically cleaned up and no longer become orphaned, eliminating the need to manually delete them.

  (DFBUGS-2827)

- **VolSync data mover pods can now access persistent volumes in environments with custom SCCs**
  Data replication operations no longer fail due to permission restrictions in clusters that enforce custom Security Context Constraints (SCCs). Previously, VolSync data mover pods lacked the necessary privileges to access underlying persistent volumes under restrictive security policies, causing synchronization to fail. This issue has been resolved by introducing a **VolSyncSpec** field to the **DRPlacementControl** (DRPC) specification, which allows users to explicitly define a custom **MoverSecurityContext** and **MoverServiceAccount** that are successfully propagated to the underlying components.

  (DFBUGS-4728)

- **Custom s3StoreProfile configuration values are no longer lost after an upgrade**
  Upgrading the OpenShift Data Foundation operator no longer causes custom configurations to be wiped from the Ramen configuration. Previously, the Multicluster Orchestrator (MCO) operator would reconcile the **ramen-hub-operator-config** ConfigMap without accounting for existing custom values, leading to the loss of the **CACertificates** and

veleroNamespaceSecretKeyRef fields under **s3StoreProfiles**. This issue has been resolved by updating the MCO operator to properly respect and preserve these custom values during upgrades.

(DFBUGS-440)

6.3. CSI DRIVER

- **Sync no longer stops after PVC deselection**
  Sync operations now continue to function correctly when a PersistentVolumeClaim (PVC) is added to or removed from a group by modifying its label. Stale protected PVC entries are now properly cleaned up from the VolumeReplicationGroup (VRG) status, eliminating the need to manually edit the VRG's status field as a workaround.

  (DFBUGS-4012)

6.4. CEPH

- **OSD disks are now properly zapped and cleaned during reinstallation**
  Disk zapping and cleanup logic now runs successfully during reinstallation even if the initial disk metadata offsets have been modified or cleared by external automation. Previously, the cleanup process relied on a mechanism that only checked the first location for BlueStore metadata; if it was missing, the process skipped zapping the rest of the disk, leaving remaining metadata intact. This issue has been resolved by migrating the disk cleanup logic directly into Rook and removing dependencies on external zapping commands.

  (DFBUGS-5549)

- **Ceph monitors can now successfully form a quorum during a network partition between data centers**
  A network split between data sites no longer causes Ceph commands and I/O operations to hang indefinitely. Previously, when the two main data centers lost connectivity with each other but maintained communication with the Arbiter node, a flaw in the monitor election logic triggered an endless loop of leader elections. This rendered the Ceph cluster unavailable and required administrators to manually shut down monitors in one data center and reset connection scores as a workaround. This issue has been resolved by fixing the election logic to handle asymmetric network disruptions correctly.

  (DFBUGS-425)

6.5. OPENSHIFT DATA FOUNDATION CONSOLE

- **UI no longer shows "Unauthorized" error and temporary blank loading screen during ODF operator installation**
  During the OpenShift Data Foundation operator installation, the page now correctly handles instances where the InstallPlan transiently goes missing. The UI no longer displays an unknown status, "Unauthorized" error, or a temporary blank screen with missing titles and messages.

  (DFBUGS-3574)

- **OpenShift Data Foundation console no longer incorrectly displays internal clusters as external**
  The OpenShift Data Foundation web console now accurately identifies and displays the cluster deployment mode. Previously, the console determined whether a cluster was internal or external

based solely on the presence of the **externalStorage.enable** field rather than its actual boolean value, occasionally causing internal clusters to be misidentified as external. This issue has been resolved by verifying the exact boolean state of the field.

(DFBUGS-6242)

6.6. OCS OPERATOR

- **The kube-apiserver no longer logs 404 error messages originating from OpenShift Data Foundation**
  The **kube-apiserver** no longer generates frequent **HTTP 404 (Not Found)** error logs related to OpenShift Data Foundation internal requests or endpoints. This issue has been resolved by correcting the underlying missing or misconfigured resource requests between OpenShift Data Foundation components and the Kubernetes API server.

  (DFBUGS-784)

6.7. UPGRADE

- **odf-blackbox-exporter deployment no longer fails when upgrading**
  The **odf-blackbox-exporter** deployment now upgrades successfully without failure. Previously, when upgrading to certain OpenShift Data Foundation versions, the deployment would fail, requiring administrators to manually delete the deployment as a workaround. This issue has been resolved.

  (DFBUGS-6789)

- **Encrypted OSDs are no longer unnecessarily redeployed during a 4.19.z upgrade**
  Upgrading legacy clusters (originally deployed on version 4.10 or earlier) to version 4.19 no longer triggers unnecessary OSD migrations and data movement. Previously, a checking mechanism failed to account for the legacy **spec.encryption.enable** setting, causing already-encrypted OSDs to be treated as unencrypted and migrated sequentially. This issue has been resolved by properly respecting the legacy encryption setting.

  (DFBUGS-6696)

- **External clusters no longer upgrade OpenShift Data Foundation automatically without manual approval**
  When upgrading OpenShift Container Platform, external OpenShift Data Foundation clusters now correctly respect manual approval settings for installation plans. Previously, the **ocs-operator** did not set the **disableInstallPlanAutoApproval** key in the **ocs-client-operator-config** ConfigMap for external deployments, which caused the client-operator to automatically approve installation plans unexpectedly. This issue has been resolved by ensuring the key is set across all cluster types.

  (DFBUGS-6043)

CHAPTER 7. KNOWN ISSUES

This section describes the known issues in Red Hat OpenShift Data Foundation 4.22.

7.1. DISASTER RECOVERY

- **DRCluster validation can fail after upgrading to 4.22 when configured CIDRs are not detected by CSINodeAddons**
  In 4.22, CIDR validation for MDR fencing has been enhanced. The CIDRs configured in **DRCluster.Spec.CIDRs** are now validated against the CIDRs reported by **CSINodeAddons** on the cluster. As a result, any CIDRs that are not detected by **CSINodeAddons** cause DRCluster validation to fail with the following condition:

  > DRClusterValidated: False — undetected CIDRs specified <cidr1>, <cidr2>

  CSINodeAddons reports the node IPs that are visible to the storage cluster. If **DRCluster.Spec.CIDRs** was configured by collecting CIDRs from all cluster nodes before upgrading, control plane node CIDRs might be present in the configuration. Because control plane nodes are typically not part of the storage network, their IPs are not reported by **CSINodeAddons** and can cause validation to fail after upgrading to 4.22.

  Workaround: Verify that the CIDRs configured in **DRCluster.Spec.CIDRs** match the CIDRs detected by the cluster, either before or after upgrading to 4.22. Post-upgrade verification is recommended because the detected CIDRs are available directly.

  Before upgrading to 4.22:

  - Verify the CIDRs configured in **DRCluster.Spec.CIDRs** against the schedulable nodes on the managed cluster.
  - Include worker nodes and any schedulable control plane nodes in the verification.
  - Correct any mismatches before upgrading.
  - Because CIDR detection is not available in 4.21, post-upgrade validation is still recommended.
    After upgrading to 4.22 (recommended):
  - Configure the required **NetworkFenceClass** resources.
  - Compare the detected CIDRs with the values configured in **DRCluster.Spec.CIDRs** and correct any mismatches.
  - Verify that the **DRClusterValidated** condition reports **Status: True**.
    For instructions on retrieving detected CIDRs and updating the DRCluster configuration, see Configure DRClusters for fencing automation .

    (DFBUGS-8068)

- **UI cleanup message identifies the wrong cluster after aborting a discovered application dryRun test failover**
  When a dryRun test failover for a discovered application is aborted, the **failoverCluster** field is removed from the DRPlacementControl (DRPC) specification while the DRPC **status.phase** remains **FailedOver**. The UI derives the cleanup target cluster by finding the complement of the

primary cluster, but since **getPrimaryClusterName()** returns undefined (due to missing **spec.failoverCluster**), the cluster lookup defaults to the first cluster in the DRPolicy - which may be the original primary cluster instead of the actual failover target.

As a result, the UI cleanup message incorrectly directs the user to delete application resources from the original primary cluster instead of the failover target cluster where the dryRun workload was deployed. Acting on this incorrect guidance could result in deleting the production workload.

Workaround: Ignore the cluster name displayed in the UI cleanup message. Instead, identify the failover target cluster to which the dryRun test failover was triggered and delete the application resources from that cluster.

Result: After cleaning up the workload from the correct failover target cluster, the application will remain running only on the original primary managed cluster and data replication will resume, restoring the DR state for the application.

(DFBUGS-7998

- **UI does not reflect DRPC progression updates during ApplicationSet dryRun test failover abort**
  When a dryRun test failover for an ApplicationSet workload is aborted, the DRPC progression correctly transitions through **Cleaning Up** and then **Completed**. However, the user interface does not recognize or display these progression updates during the abort workflow.

  As a result, the UI continues to display a stale state, such as **TestingFailover** or **FailedOver**, instead of the actual DRPC progression. This can give users an inaccurate view of the dryRun abort and cleanup status.

  Workaround: Verify the actual DRPC progression by using the command line instead of relying on the UI status. For example, run **oc get drpc <name> -n <namespace> -o yaml** and review the DRPC progression details to confirm that the abort and cleanup have completed successfully.

  Result: After the dryRun abort completes and the DRPC progression reaches **Completed**, the workload is removed from the failover cluster and normal disaster recovery protection resumes on the primary managed cluster.

  (DFBUGS-8070)

- **CephFS VolSync PVCs can get stuck during disable DR**
  During disable DR for CephFS workloads protected by VolSync, Ramen deletes **ReplicationSource** (and related replication-group) objects without first removing their owner references from the PVCs, even when the disable-DR workflow requests that PVCs be preserved (**do-not-delete-pvc**). Kubernetes then marks the PVCs for deletion.

  As a result, **PersistentVolumeClaims** can remain stuck in a **Terminating** state instead of staying **Bound**. Disable DR does not complete cleanly for affected CephFS deployments, and workloads might continue running while storage is left in an inconsistent lifecycle state.

  Workaround: Do not disable DR for CephFS VolSync workloads until a fix is available.

  If PVCs are already stuck in a **Terminating** state, recovery requires manual intervention and is not suitable for customer self-service. One manual recovery method is to use **kubectl patch** to remove the **ReplicationSource** and **ReplicationDestination** owner references from the PVCs during the disable-DR operation.

(DFBUGS-8196)

- **Ramen fails to detect updated SSL certificates for S3 connections**
After an SSL certificate rotation on the S3 service, Ramen might not automatically detect newly added SSL certificates in the hub cluster. The Ramen pod continues to use the certificates that were loaded during startup, which can cause S3 connections to newly added clusters to fail.

  This issue might surface as errors such as:

  ```
  failed to list objects in bucket: code: RequestError, message: send request failed
  ```

  Note that this error can occur due to various connection or access issues and should not always be interpreted as an SSL certificate problem.

  Workaround: Manually restart the Ramen pod on the hub cluster to trigger reinitialization of the S3 connections. This process allows Ramen to load and use the updated SSL certificates.

  If you are using managed clusters:

- Update the certificates in the **managedClusters** configuration.
- Manually restart the Ramen pods in the managed clusters after the certificate rotation is complete.
(DFBUGS-2520)

- **CephBlockPoolRadosNamespace remains in Progressing state for custom CephBlockPools in an RDR configuration**
When a custom CephBlockPool is created on a cluster configured for Regional Disaster Recovery (RDR), the automatically created CephBlockPoolRadosNamespace resource (**<pool-name>-builtin-implicit**) can remain in the **Progressing** state indefinitely. This occurs because the MirroringController in the **ocs-operator** cannot enable mirroring on the CephBlockPool when the peer cluster bootstrap token is not yet available.

  As a result, mirroring is not enabled on the CephBlockPool and the associated CephBlockPoolRadosNamespace does not transition to a healthy state.

  Workaround: If the CephBlockPoolRadosNamespace remains in the **Progressing** state and mirroring is not enabled on the CephBlockPool, restart the **ocs-operator** pod.

  Result: After the **ocs-operator** restarts, mirroring is enabled successfully on the CephBlockPool and the CephBlockPoolRadosNamespace transitions out of the **Progressing** state.

  (DFBUGS-7981)

- **DRPCs protect all persistent volume claims created on the same namespace**
The namespaces that host multiple disaster recovery (DR) protected workloads protect all the persistent volume claims (PVCs) within the namespace for each DRPlacementControl resource in the same namespace on the hub cluster that does not specify and isolate PVCs based on the workload using its **spec.pvcSelector** field.

  This results in PVCs that match the DRPlacementControl **spec.pvcSelector** across multiple workloads. Or, if the selector is missing across all workloads, replication management to potentially manage each PVC multiple times and cause data corruption or invalid operations based on individual DRPlacementControl actions.

  Workaround: Label PVCs that belong to a workload uniquely, and use the selected label as the

DRPlacementControl **spec.pvcSelector** to disambiguate which DRPlacementControl protects and manages which subset of PVCs within a namespace. It is not possible to specify the **spec.pvcSelector** field for the DRPlacementControl using the user interface, hence the DRPlacementControl for such applications must be deleted and created using the command line.

Result: PVCs are no longer managed by multiple DRPlacementControl resources and do not cause any operation and data inconsistencies.

(DFBUGS-1749)

- **Disabled PeerReady flag prevents changing the action to Failover**
  The DR controller executes full reconciliation as and when needed. When a cluster becomes inaccessible, the DR controller performs a sanity check. If the workload is already relocated, this sanity check causes the **PeerReady** flag associated with the workload to be disabled, and the sanity check does not complete due to the cluster being offline. As a result, the disabled **PeerReady** flag prevents you from changing the action to Failover.

  Workaround: Use the command-line interface to change the DR action to Failover despite the disabled **PeerReady** flag.

  (DFBUGS-665)

- **For discovered apps with CephFS, sync stop after failover**
  For CephFS-based workloads, synchronization of discovered applications may stop at some point after a failover or relocation. This can occur with a **Permission Denied** error reported in the **ReplicationSource** status.

  Workaround:

  - **For Non-Discovered Applications**
    - Delete the VolumeSnapshot:

      ```
      $ oc delete volumesnapshot -n <vrg-namespace> <volumesnapshot-name>
      ```

      The snapshot name usually starts with the PVC name followed by a timestamp.

    - Delete the VolSync Job:

      ```
      $ oc delete job -n <vrg-namespace> <pvc-name>
      ```

      The job name matches the PVC name.

  - **For Discovered Applications**
    Use the same steps as above, except **<namespace>** refers to the **application workload namespace**, not the VRG namespace.

  - **For Workloads Using Consistency Groups**
    - Delete the ReplicationGroupSource:

      ```
      $ oc delete replicationgroupsource -n <namespace> <name>
      ```

    - Delete All VolSync Jobs in that Namespace:

```
$ oc delete jobs --all -n <namespace>
```

In this case, **<namespace>** refers to the namespace of the workload (either discovered or not), and **<name>** refers to the name of the ReplicationGroupSource resource.

(DFBUGS-2883)

- **DRPC progression completes without workload cleanup when aborting a CephFS dryRun test failover**
  For a CephFS discovered application that is in the **Deployed** state before a dryRun test failover is triggered, aborting the dryRun can cause the DRPlacementControl (DRPC) progression to transition from **TestingFailover** to **Completed** even though the workload has not been cleaned up from the failover cluster (test cluster).

  This results in the CephFS discovered application continuing to run on both managed clusters simultaneously, which is not a desirable state and can leave the application's disaster recovery state inconsistent.

  Workaround: Delete the application from the failover cluster (test cluster) where the dryRun test failover was triggered.

  Result: After the application is successfully removed from the failover cluster, the workload continues running only on the primary managed cluster. Data replication resumes after the cleanup is completed, and the application's disaster recovery state is restored.

  (DFBUGS-8115)

7.2. MULTICLOUD OBJECT GATEWAY

- **Unable to create new OBCs using Multicloud Object Gateway**
  When provisioning an NSFS bucket via ObjectBucketClaim (OBC), the default filesystem path is expected to use the bucket name. However, if path is set in **OBC.Spec.AdditionalConfig**, it should take precedence. This behavior is currently inconsistent, resulting in failures when creating new OBCs.

  (DFBUGS-3817)

7.3. CEPH

- **OSD pods restart during add capacity**
  OSD pods restart after performing cluster expansion by adding capacity to the cluster. However, no impact to the cluster is observed apart from pod restarting.

  (DFBUGS-1426)

- **SELinux relabelling issue with a very high number of files**
  When attaching volumes to pods in Red Hat OpenShift Container Platform, the pods sometimes do not start or take an excessive amount of time to start. This behavior is generic and it is tied to how SELinux relabelling is handled by Kubelet. This issue is observed with any filesystem based volumes having very high file counts. In OpenShift Data Foundation, the issue is seen when using CephFS based volumes with a very high number of files. There are multiple ways to work around this issue. Depending on your business needs you can choose one of the workarounds from the knowledgebase solution https://access.redhat.com/solutions/6221251.

  (RFE-3327)

7.4. OPENSHIFT DATA FOUNDATION CONSOLE

- **UI shows WaitOnUserCleanUp even when automatic cleanup is enabled**
  The UI incorrectly displays the **WaitOnUserCleanUp** status even when automatic cleanup is enabled for VMs. This occurs because the UI relies only on the **phase** and **progression** fields of the **DRPlacementControl** to determine cleanup behavior and does not evaluate the more granular **AutoCleanup** condition that explicitly indicates automatic cleanup.

  Workaround: There is no manual workaround required. This state is transient and clears automatically once the **progression** field advances to **Completed**. Manual cleanup should be avoided unless the **AutoCleanup** condition and its corresponding **reason** in the **DRPlacementControl** or VRG status indicate otherwise.

  During automatic cleanup, the UI may briefly present a misleading status, which can cause temporary confusion until the cleanup completes.

  (DFBUGS-5824)

- **DRPlacementControl shows ProtectionError even after successful relocation**
  When a relocation completes, the **DRPlacementControl** may continue to display a **ProtectionError** status. This occurs because the **Protected** condition in the DRPlacementControl status incorrectly reports an **Error** state, even though the relocation has finished (**phase: Relocated**, **progression: Completed**).

  Workaround: No direct workaround is available. Wait until retrying the **NoClusterDataConflict** condition is met.

  The DR status in the UI remains in the **ProtectionError** state until the data conflict is resolved.

  (DFBUGS-5823)

CHAPTER 8. DEPRECATED FEATURES

This section describes the deprecated features introduced in Red Hat OpenShift Data foundation 4.22.

8.1. DEPRECATION NOTICE: METRO DR FOR EXTERNAL MODE ARCHITECTURES

Metro DR (Disaster Recovery) functionality based on external OpenShift Data Foundation and Red Hat Ceph Storage architectures is deprecated as of this release (4.22) and will be removed in version 5.1. OpenShift Data Foundation 4.23 will be the final release to support this configuration.

Starting with version 5.1, future Metro DR capabilities will transition exclusively to a Provider-mode architecture within IBM Storage Fusion Data Foundation. OpenShift Data Foundation will no longer natively provide standalone Metro DR capabilities.

Impact and Migration Path:

- **Support Lifecycle:** Existing customers utilizing this technology can safely remain on version 4.23, which will receive full support until its End of Life (EOL) at the end of 2029.
- **Next Steps:** Customers planning future deployments or looking to upgrade beyond the 4.x lifecycle are encouraged to plan a migration to the new Provider-mode Metro DR architecture on IBM Storage Fusion Data Foundation.